전산화학 (2013년 1학기) 1차 과제 해답

(총 50점)

1. **(총 20점, 각 10점)** 그래프는 맨 뒤에.

(a)

$$f(x) = \begin{cases} 0 & (-L \le x < 0) \\ x & (0 \le x < L) \end{cases}$$

Fourier 급수로 나타내기 위하여

$$\begin{aligned} a_n &= \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx, \\ &= \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)dx, \\ &= \frac{1}{L}\left[\frac{L}{n\pi}x\sin\left(\frac{n\pi x}{L}\right)\right]_0^L - \frac{1}{L}\int_0^L \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)dx \quad \text{부분적분 이용} \\ &= \frac{1}{n\pi}\left[\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_0^L \\ &= \frac{L}{(n\pi)^2}\big(\cos(n\pi) - 1\big) \\ &= \frac{L}{(n\pi)^2}\big((-1)^n - 1\big) \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx, \\ &= \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)dx, \\ &= \frac{1}{L}\left[-\frac{L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)\right]_0^L + \frac{1}{L}\int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)dx \quad \text{부분적분 이용} \\ &= -\frac{L}{n\pi}\cos(n\pi) + \frac{1}{n\pi}\left[\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\right]_0^L \\ &= -\frac{L}{n\pi}(-1)^n \\ &= \frac{L}{n\pi}(-1)^{n+1}. \end{aligned}$$

한편,

$$\begin{aligned} a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)dx \\ &= \frac{1}{L}\int_0^L x dx \\ &= \frac{1}{L}\left[\frac{1}{2}x^2\right]_0^L x dx \\ &= \frac{L}{2} \end{aligned}$$

그러므로 Fourier 급수는 다음과 같이 주어진다.

$$\boxed{f(x) = \frac{L}{4} + \sum_{n=1}^{\infty} \frac{L}{(n\pi)^2}\left((-1)^n - 1\right)\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} \frac{L}{n\pi}(-1)^{n+1}\sin\left(\frac{n\pi x}{L}\right).} \tag{1}$$

다음과 같은 그래프를 얻을 수 있다.

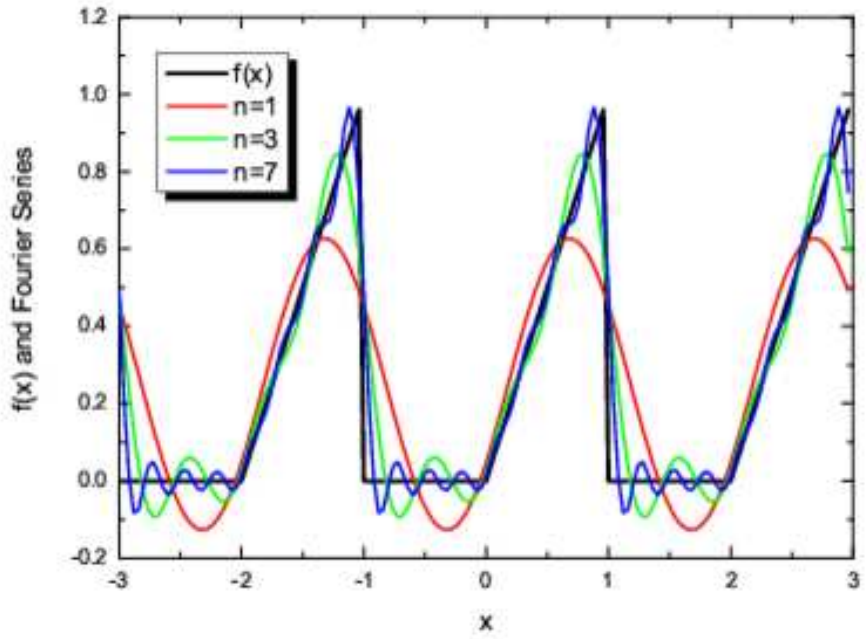


그림 1: 과제 1번 (a)

(b)

$$f(x) = \begin{cases} x(L+x) & (-L \le x < 0) \\ x(L-x) & (0 \le x < L) \end{cases}$$

Fourier 급수의 계수를 구하면 a_n 은

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx,$$

$$= 0 \qquad (\ f(x) \text{는 기함수이므로})$$

b_n 은

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx,$$

$f(x)$ 는 기함수이고 $\sin(n\pi x/L)$ 도 기함수이므로 $f(x)\sin(n\pi x/L)$ 는 우함수, 그러므로

$$= \frac{2}{L}\int_{0}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx,$$

$$
\begin{aligned}
&= \frac{2}{L}\int_0^L x(L-x)\sin\left(\frac{n\pi x}{L}\right)dx, \\
&= -\frac{2}{L}\left[\frac{L}{n\pi}x(L-x)\cos\left(\frac{n\pi x}{L}\right)\right]_0^L + \frac{2}{L}\int_0^L \frac{L}{n\pi}(L-2x)\cos\left(\frac{n\pi x}{L}\right)dx \quad \text{(부분적분 이용)} \\
&= \frac{2}{n\pi}\int_0^L (L-2x)\cos\left(\frac{n\pi x}{L}\right)dx \\
&= \frac{2}{n\pi}\left[\frac{L}{n\pi}(L-2x)\sin\left(\frac{n\pi x}{L}\right)\right]_0^L - \frac{2}{n\pi}\int_0^L \frac{L}{n\pi}(-2)\sin\left(\frac{n\pi x}{L}\right)dx \quad \text{(부분적분 이용)} \\
&= \frac{4L}{(n\pi)^2}\int_0^L \sin\left(\frac{n\pi x}{L}\right)dx \\
&= \frac{4L}{(n\pi)^2}\left[-\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\right]_0^L \\
&= -\frac{4L^2}{(n\pi)^3}\big(\cos(n\pi)-1\big) \\
&= -\frac{4L^2}{(n\pi)^3}\big((-1)^n-1\big) = \frac{4L^2}{(n\pi)^3}\big(1-(-1)^n\big)
\end{aligned}
$$

그러므로 Fourier 급수는 다음과 같이 주어진다.

$$
\boxed{f(x) = \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^3}\big(1-(-1)^n\big)\sin\left(\frac{n\pi x}{L}\right),} \tag{2}
$$

다음과 같은 그래프를 얻을 수 있다.

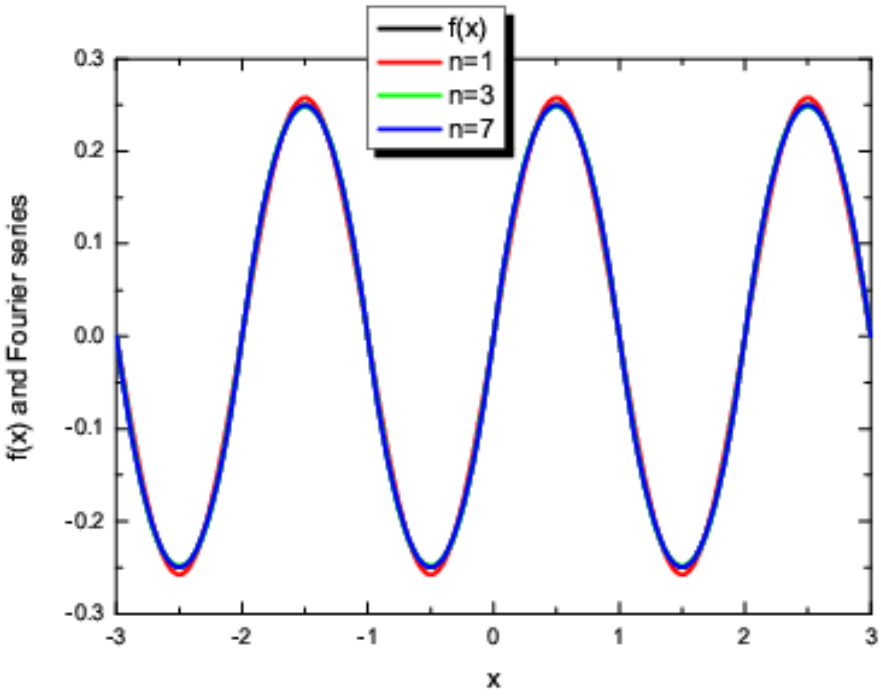


그림 2: 과제 1번 (b)

2. **(총 20점, 각 10점)**

(a) 문제 1의 (a)에서 구한 (1) 식에 $x = 0$을 대입하면

$$f(0) = 0 = \frac{L}{4} + \sum_{n=1}^{\infty} \frac{L}{(n\pi)^2}\left((-1)^n - 1\right)$$

위 식을 전개하면

$$= \frac{L}{4} + \frac{L}{\pi^2}(-2) + \frac{L}{(3\pi)^2}(-2) + \frac{L}{(5\pi)^2} + \cdots$$
$$\Rightarrow 0 = \frac{L}{4} - \frac{2L}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right)$$

정리하면

$$\boxed{1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}}$$

(b) 문제 1의 (b)에서 구한 (2) 식에 $x = L/2$을 대입하면

$$f(L/2) = \frac{L^2}{4} = \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^3}\left(1 - (-1)^n\right)\sin\left(\frac{n\pi L}{2L}\right),$$
$$= \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^3}\left(1 - (-1)^n\right)\sin\left(\frac{n\pi}{2}\right),$$

n에 대해 전개하면

$$= \frac{4L^2}{\pi^3}\left(\frac{2}{1^3} - \frac{2}{3^3} + \frac{2}{5^3} + \cdots + \frac{2(-1)^{n+1}}{(2n-1)^3} + \cdots\right)$$
$$= \frac{8L^2}{\pi^3}\left(\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} + \cdots + \frac{(-1)^{n+1}}{(2n-1)^3} + \cdots\right)$$

그러므로

$$\frac{L^2}{4} = \frac{8L^2}{\pi^3}\left(\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} + \cdots + \frac{(-1)^{n+1}}{(2n-1)^3} + \cdots\right)$$

이고,

$$\boxed{\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} + \cdots + \frac{(-1)^{n+1}}{(2n-1)^3} + \cdots = \frac{\pi^3}{32}}$$

3. **(10점)**

$$
\begin{aligned}
g(\omega) &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \\
&= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} e^{-\alpha|t|} \cos(\omega_0 t) e^{-i\omega t} dt \\
&= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} \underbrace{e^{-\alpha|t|} \cos(\omega_0 t)}_{\text{우함수}} \underbrace{\cos(\omega t)}_{\text{우함수}} dt - i \int_{-\infty}^{\infty} \underbrace{e^{-\alpha|t|} \cos(\omega_0 t)}_{\text{우함수}} \underbrace{\sin(\omega t)}_{\text{기함수}} dt
\end{aligned}
$$

위 식 중에 두 번째 항은 0 0이므로

$$
\begin{aligned}
&= \frac{2}{(2\pi)^{1/2}} \int_{0}^{\infty} e^{-\alpha t} \cos(\omega_0 t) \cos(\omega t) dt \\
&= \frac{2}{(2\pi)^{1/2}} \int_{0}^{\infty} e^{-\alpha t} \frac{\cos(\omega_0 + \omega)t + \cos(\omega_0 - \omega)t}{2} dt \\
&= \frac{1}{(2\pi)^{1/2}} \left[\int_{0}^{\infty} e^{-\alpha t} \cos(\omega_0 + \omega)t \mathrm{d}t + \int_{0}^{\infty} e^{-\alpha t} \cos(\omega_0 - \omega) \mathrm{d}t \right]
\end{aligned}
$$

적분 공식에 의해 위 두 항의 적분은 각각

$$
= \frac{1}{(2\pi)^{1/2}} \left[\frac{\alpha}{(\omega_0 + \omega)^2 + \alpha^2} + \frac{\alpha}{(\omega_0 - \omega)^2 + \alpha^2} \right]
$$